

Domain Adaptive Object Detection (DAOD)

Traditional offline Unsupervised Domain Adaptation (UDA) is rapidly being superseded by privacy-preserving, online, and foundation-model-driven approaches that operate without access to source data.

Adaptation Paradigm	Description and Operational Context	Source Data Access	Target Data Access	Real-Time Execution
Unsupervised Domain Adaptation (UDA)	The foundational offline adaptation paradigm. It transfers knowledge from a fully labelled source dataset to a completely unlabelled target dataset by simultaneously processing both distributions to find invariant features.	Yes (Full Access)	Yes (Unlabelled)	No
Semi-Supervised Domain Adaptation (SSDA)	A hybrid approach that utilises full source data alongside a very small fraction of labelled target data and a large pool of unlabelled target data. This is common when minimal target annotation budget is available.	Yes (Full Access)	Yes (Sparse Labels)	No
Source-Free Domain Adaptation (SFDA)	Adapts a pre-trained source model to a target domain <u>without ever accessing the original source data</u> . This is currently a major research focus due to stringent commercial data privacy regulations and medical confidentiality.	No	Yes (Unlabelled)	No
Test-Time Adaptation (TTA)	An online adaptation strategy that <u>adjusts the model's parameters on-the-fly during inference</u> . It uses <u>continuous streams of target data</u> to update the model in real-world, dynamic deployment scenarios.	No	Yes (Streaming)	Yes
Continual Test-Time Adaptation (CTTA)	An advanced extension of TTA designed to handle <u>non-stationary, constantly evolving target distributions</u> (e.g., <u>transitioning from day to night to rain</u>). It specifically addresses and <u>mitigates catastrophic forgetting of source knowledge</u> .	No	Yes (Streaming)	Yes

1- both source, target accessible

→ VDA → 2- target unlabelled

→ SS DA → both S, T available, Target sparsely labeled

adaption
Paradigm

→ SFDA (source free) → 1- source unavailable
2- target unlabelled

3- major research focus

4- privacy, confidentiality

→ TTA (test time) → 1- S unavailable, T streaming without label

2- adjust model parameters on fly during inferencing

→ cTTA (continual TTA): 1- non stationary, continual evolving target data distribution (like from day to night then to raining)

2- mitigate catastrophic forgetting of source knowledge

Alignment Strategy	Methodological Mechanism and Data Flow
<u>Discrepancy-Based Alignment</u>	These methods seek to explicitly minimise a chosen statistical distance metric between the source and target feature distributions <u>within a shared embedding space</u> . Common mathematical metrics include <u>Maximum Mean Discrepancy (MMD)</u> , <u>Correlation Alignment (CORAL)</u> , and <u>Jensen-Shannon (JS) Divergence</u> . The data flows through the <u>feature extractor</u> , the <u>discrepancy is calculated</u> , and <u>backpropagation forces the distributions to merge</u> .
<u>Adversarial Feature Learning</u>	Inspired by Generative Adversarial Networks (GANs), these architectures employ a <u>Gradient Reversal Layer (GRL)</u> and a dedicated <u>domain discriminator</u> . The feature extractor is trained to perform well on the <u>primary detection task</u> while <u>simultaneously attempting to fool the domain discriminator</u> into being unable to identify the origin domain of the features. This <u>zero-sum game yields domain-invariant representations</u> .
<u>Self-Training and Pseudo-Labelling</u>	The model generates predictions on the unlabelled target data, <u>filtering them by predefined confidence thresholds</u> to create <u>artificial 'pseudo-labels'</u> . These are typically implemented via a <u>Mean-Teacher architecture</u> , which utilises an <u>Exponential Moving Average (EMA)</u> to stabilise the <u>teacher network's weights against fluctuating student predictions</u> . The data flows from the teacher to generate labels, which supervise the student.
Image-to-Image Translation	Generative models, such as <u>CycleGAN</u> or <u>Contrastive Unpaired Translation (CUT)</u> , translate source images to mimic the target style at the pixel level (e.g., translating daytime highway scenes to nighttime scenes). This creates a synthetic, artificial dataset that perfectly retains the original source bounding box annotations, allowing for standard supervised training on the new target-like domain.
Vision-Language Model (VLM) Prompting	Representing the absolute edge of 2026 methodologies. These approaches utilise frozen multi-modal foundation models (such as Grounding DINO or CLIP). The adaptation data flow relies on aligning textual prompts with shifting visual embeddings to achieve zero-shot or highly efficient test-time adaptation, frequently bypassing the need for intensive backpropagation entirely.

discrepancy
based

- 1- minimize statistic metric of S and T in shared feature space
- 2- metrics: MMD, coral, Jensen-Shannon
- 3- feature extractor $\rightarrow$ discrepancy calculate $\rightarrow$ backpropagate $\rightarrow$ force distributions to merge

alignment strategies $\rightarrow$ adversarial $\rightarrow$ 1- inspired from GAN, employ gradient reversal layer (GRL), dedicated domain discriminator $\rightarrow$ zero sum game, domain-invariant feature representation

self training

- 1- model: Predict unlabeled target $\rightarrow$ filter by confident threshold (create pseudo label)
- 2- architecture mean teacher student + EMA (stable teacher weight against student prediction fluctuation)

img to img: generative model (cycle gen, contrastive unpaired translation (cut))

- 2- translate source img to mimic target style at pixel level
- 3- synthetic dataset to retrain $\rightarrow$ standard supervised training on new target like domain

VLLM: using multimodal models (grounding Dino, clip)

- 2- assign textual prompt with shift of visual embedding to achieve zero-shot $\rightarrow$ bypass intensive backprop on t+data

Level 2: Architectural Granularity

Unlike image classification, which condenses an image into a singular semantic meaning, object detection requires alignment at multiple granularities due to the structural complexity of identifying multiple foreground instances against massive, dominating background noise.

Granularity Level	Alignment Target and Impact
Image-Level (Global) Alignment	Aligns the <u>global context and overall scene layout</u> . This addresses <u>massive shifts</u> such as weather changes, global illumination, and <u>camera sensor differences</u> . It <u>extracts and aligns features from the entire image or the deep convolutional backbone</u> before <u>region proposals</u> are generated.
Instance-Level (Local) Alignment	Focuses strictly on aligning the <u>object proposals or Regions of Interest (ROIs)</u> . This is an absolute necessity in DAOD because <u>foreground objects (e.g., vehicles, pedestrians) suffer fundamentally different domain shifts than background elements (e.g., sky, roads)</u> . <u>Treating them equally leads to negative transfer</u> .
Pixel-Level Alignment	Operates at the lowest level of granularity, aligning low-level features such as textures, edges, and colour distributions. This is frequently achieved using the aforementioned <u>generative Image-to-Image translation techniques</u> <u>before the data even enters the detector</u> .
Query-Level Alignment	Specific to Detection Transformers (DETR). Because DETRs do <u>not use anchor boxes or RPNs</u> , adaptation occurs by <u>aligning the learnable object queries</u> . This involves <u>query active preselection</u> and <u>optimal transport matching</u> to facilitate effective <u>cross-domain knowledge transfer</u> .

img level (global align): global context, scene layout $\rightarrow$ for massive shifts (weather changes, global illumination, camera sensor change)

2- extract, align feature from entire img (CNN) $\rightarrow$ then region proposal generate

instance level (local): for obj proposal & ROI (region of interest)
2- foreground obj (car) different domain shift than background element (road, sky) $\rightarrow$ ex. transfer lead negative transfer

pixel level: 1- lowest level, like texture, edge, color distribution 2- use generative img2img before data get in detector

Query level: DETR model, not use anchor boxes or RPN
2- align learnable obj queries
3- involve query pre selection, optimal transport matching to facilitate cross-domain knowledge transfer